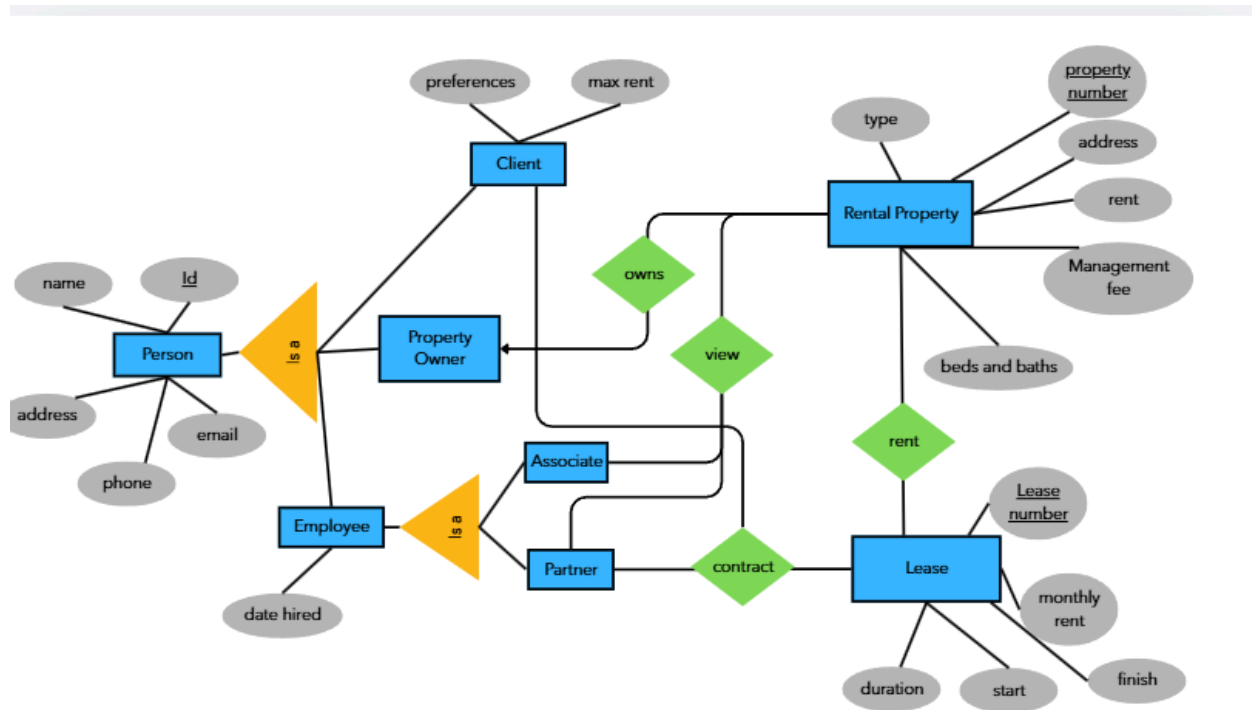


Phase C:

For this phase, the conceptual design for the database from Part B is modeled by this ER diagram:



To map this design into a relational data model, a series of reductions on each of the entities and relations were performed to change this ER model to a relational schema.

- Person (id, name, address, phone, email)
 - Pk: id
 - Fk: none

This is the main entity representing all people in the database. It groups common attributes shared across all roles among people to avoiding redundancy

- Employee (employee id, hired_date)
 - Pk: employee id
 - Fk: employee_id → Person(id)

Specialization of person that adds the hired_date attribute. This is a separate table where an employee inheriting their values in person is implemented via foreign key reference

- Property owner (id)
 - Pk: id
 - Fk: id → Person(id)

Specialization of person entity. Has relations specific to this kind of entity so this was made into its own table and is linked to the person table through a foreign key reference

- Client (id, property preferences, max_rent)
 - Pk: id
 - Fk: id → Person(id)

Specialization of Person. Adds the property preference and max rent attributes. Has its own specific type of relations compared to a regular person

- Partner (employee_id)
 - Pk: employee id
 - Fk: employee_id → Employee(employee_id)

Specialization of an employee that has the ability to lease a contract unlike an associate. It is separated to enforce role-specific constraints and relationships

- Associate(employee_id)
 - Pk: employee id
 - Fk: employee_id → Employee(employee_id)

Specialization of employees. Relation created for an Employee representing employees responsible for assisting clients

- Rental Property(property_number, type, address, rent, management_fee, bed_baths)
 - Pk: property_number
 - Fk: none

This entity represents properties available for rent. This relation is its own base entity.

- Lease (lease number, property_number, rent, deposit, duration, start, finish)
 - Pk: lease_number, property_number
 - Fk: property_number → RentalProperty(property_number)

This lease relation identifies a lease tied to a specific property. A composite primary key is used to ensure that a lease is uniquely identified within the context of a property

- Contract(partner_id, lease_number, client_id, property_number)
 - Pk: partner_id, lease_number, property_number, client_id
 - Fk: partner_id → partner(employee_id)
 - client_id → client(id)
 - property_number → RentalProperty(property_number)

This relation models the ternary relationship between Partner, Client, and Lease. It captures which partner arranged a lease for which client and property.

- View(client_id, associate_id, property_number)
 - Pk: client_id, associate_id, property_number
 - Fk: client_id → Client(id)
 - associate_id → Associate(employee_id)
 - property_number → RentalProperty(property_number)
- Owns(owner_id, property_number)
 - PK: owner_id, property_number
 - FK: owner_id → PropertyOwner(id)
 - property_number → RentalProperty(property_number)

The rent relation is excluded from this relation as it would be represented as the identifying keys of the Rental Property and Lease entities, however this is already captured by the Lease entity as its primary key is (lease_number, property_number)

Phase D:

For this phase, I converted the relations from Phase C into SQL tables in order to physically create my conceptual database. The tables from the relations in Phase C follow for the most part, the exact same information as the tuple above.

